[2CEIT401 – Operating Systems]

Practical: 3

AIM- Implements Shell script using decision making statement, branching and Looping.

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Aim: Implements Shell script using decision making statement, branching and Looping.

1. Write a shell script to calculate the loss percentage of an article. Scan the cost price and selling price

Ans:

Code:

```
echo "Enter Cost Price: "
read cost_price
echo "Enter Selling Price: "
read selling_price
loss=`expr $cost_price - $selling_price`
loss=`expr $loss \* 100`
loss_percent=`expr $loss / $cost_price`
loss_percent1=`echo "scale=2; $loss / $cost_price" | bc`
echo "Loss Percentage: $loss_percent1% By bc"
echo "Loss Percentage: $loss_percent% By expr"
```

Output:

```
[24012011142@linuxserv prac3]$ sh que_1.sh
Enter Cost Price:
100
Enter Selling Price:
95
loss: 5.00% By bc
Loss: 5% by expr
```

2. Write a shell script to accept numbers below 50 and to display the square of each. This should continue as long as the user wishes.

Ans:

Code:

```
isRunning=1

while [ $isRunning -eq 1 ]

do

    echo "Enter Number less Then 50: "

    read num
```

```
    while [ $num -ge 50 ]
    do
        echo "Please Enter Number less Then 50: "
        read num
    done
    square=`expr $num \* $num`
    echo "Square of $num is: $square"
    echo "Do you want to continue? (y,n): "
    read ch
    if [ $ch = "n" ] || [ $ch = "N" ]
    then
        isRunning=0
    fi
done
```

Output:

```
[24012011142@linuxserv prac3]$ sh que_2.sh
Enter Number Less Then 50:
90
Please Enter Number less Then 50:
20
Square of  is: 400
Do you want to continue? (y/n):
y
Enter Number Less Then 50:
25
Square of  is: 625
Do you want to continue? (y/n):
n
[24012011142@linuxserv prac3]$
```

3. Write a shell script to display $50

Ans:

Code:

```
echo \$50
```

Output:

```
[24012011142@linuxserv prac3]$ sh que_3.sh
$50
```

4. Write a shell script to check whether the scanned string is found in a file or not. Display appropriate message.

Ans:

Code:

```
a=`grep $1 $2`
if [ $? -eq 0 ]
then
   echo "String Found"
else
   echo "String Not Found"
fi
```

Output:

```
[24012011142@linuxserv prac3]$ sh que_4.sh UVPCE abc.txt
String Found
[24012011142@linuxserv prac3]$ sh que_4.sh BSPP abc.txt
String Not Found
```

5. Write a shell script, which scans the name of the command and executes it.

Code:

```
echo "Enter Name of Command You Want to Run: "

read cammand

$cammand
```

Output:

```
[24012011142@linuxserv prac3]$ sh que_5.sh
Enter Name of Command You Want to Run:
ls -l
total 36
-rw-rw-r-- 1 24012011142 24012011142   6 Jan 29 18:44 abc.txt
-rw-rw-r-- 1 24012011142 24012011142 393 Feb 12 04:00 que_1.sh
-rw-rw-r-- 1 24012011142 24012011142 360 Feb 12 04:07 que_2.sh
-rw-rw-r-- 1 24012011142 24012011142  10 Feb 12 04:09 que_3.sh
-rw-rw-r-- 1 24012011142 24012011142  90 Feb 12 04:12 que_4.sh
-rw-rw-r-- 1 24012011142 24012011142  68 Feb 12 04:16 que_5.sh
-rw-rw-r-- 1 24012011142 24012011142  92 Feb  1 01:54 que_6.sh
-rw-rw-r-- 1 24012011142 24012011142 256 Feb  1 02:15 que_7.sh
-rw-rw-r-- 1 24012011142 24012011142  39 Feb  1 02:04 temp.sh
```

6. Write a shell script which displays January if we enter Jan, Janu, Janua or January.

Code:

```
echo "Enter Month: "

read month

case $month in

Jan|Janu|Janua|Januar|January)

echo "January";;

*) echo "Invalid Input";;

esac
```

Output1:

```
[24012011142@linuxserv prac3]$ sh que_6.sh
Enter String
Jan
January
```

7. Write a shell script to generate Fibonacci series.

Code:

```
echo "Enter Number 1"
read num1
echo "Enter Number 2"
read num2
echo "How Many Fibonnaci Numbers You Want to Print"
read n
n=`expr $n - 2`
i=0
fib=("$num1" "$num2")
while [ $n -gt $i ]
do
    idx=`expr $i + 2`
    temp=`expr ${fib[$i]} + ${fib[$i+1]}`
    fib[$idx]=$temp
    i=`expr $i + 1`
done
echo "Fibonnaci Series: ${fib[@]}"
```

Output:

```
[24012011142@linuxserv prac3]$ sh que_7.sh
Enter Number 1
2
Enter Number 2
5
How Many Fibonnaci Number You Want to Print?
8
Fibonnaci Series: 2 5 7 12 19 31 50 81
```